

Arithmetic and complexity

Exact bilinear rank of the 3×3 matrix product

Difficulty: extreme **Importance:** interesting to the community**Status:** Open

Rating rationale: Extreme because the exact rank of this small multiplication tensor remains a decades-old barrier despite extensive algorithm searches; community importance concerns recursive matrix multiplication and bilinear algorithm design.

Topic: small matrix multiplication algorithms

Problem statement

Determine the least integer r for which there are complex coefficients $u_{t,ij}, v_{t,ij}, w_{ij,t}$ satisfying, for every pair of complex 3×3 matrices,

$$(AB)_{ij} = \sum_{t=1}^r w_{ij,t} \left(\sum_{a,b=1}^3 u_{t,ab} A_{ab} \right) \left(\sum_{c,d=1}^3 v_{t,cd} B_{cd} \right).$$

Scalar linear combinations are free in this bilinear-rank model. The classical upper bound is 23; deciding whether 22 products suffice is part of determining the exact minimum. Algorithms mixing entries of both inputs within a factor are outside this specified model.

Why it matters

Such identities can be applied recursively to matrix blocks.

References and status

Bläser, 2013, §1 and Example 5.10. Y. Sun, *An Exact 56-Addition, Rank-23 Scheme for General 3×3 Matrix Multiplication* (2026), gives another 23-product scheme. C. Wang, *Automated Lower Bounds for Bilinear Complexity over Finite Fields*, v10 (2026), improves a lower bound over $\mathbb{F}_2$. Rudich and Rousseau’s October preprint, discussed below, raises the finite-field lower bound to 22; it does not establish the same lower bound over $\mathbb{C}$. Searches for complex bilinear algorithms and matching lower bounds found no determination of the exact complex rank. **Admitted: no resolution located.**

Status check — 2026-09-10

Rechecked [Sun’s rank-23 construction](#) and [Alman–Li, §1](#), which explicitly identifies both exact rank and border rank of the 3×3 product as unresolved. Searches for 22-product complex bilinear algorithms found no qualifying construction or matching lower bound. Fewer additions at rank 23 and finite-field lower bounds do not settle this complex-field minimum.

Reviewed research submission — 14 September 2026

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Reviewed results cover fixed-support calculations, the Sun four-block conditional obstruction and two five-block examples. Unrestricted complex rank remains undetermined. Missing continuation checkers prevent certification of its larger noncompletion claims; these are excluded from the passing audit.

Status: Open. [Report](#) · [Independent AI-agent audit and limitations](#). AI assistance disclosed; no external human review, formal verification or Lean checks. Original target and prior-source credit retained.

Literature update — 2026-10-06

Isaac Rudich and Louis-Martin Rousseau, *Lower Bound of 22 for 3x3 Matrix Multiplication over the Integers*, arXiv:2610.01639v1, submitted 1 October 2026, reports in §1 that the bilinear rank of the 3×3 matrix product over $\mathbb{F}_2$ is at least 22. §3, [displayed theorem \(1\)](#), *recursive_algorithm_multiplications_at_least_22*, gives the corresponding lower bound for algorithms with integer constants that work on square matrix blocks of every size. Sections 3–4 specify this model and the reduction from integer-coefficient algorithms to decompositions over $\mathbb{F}_2$.

This advances the finite-field and integer-coefficient problems, where the lower bound 22 and upper bound 23 still differ. It does not establish a lower bound of 22 over $\mathbb{C}$: arbitrary complex coefficients cannot in general be reduced modulo 2. The exact complex bilinear rank asked for here remains undetermined, and **Status: Open** is retained. The authors supply a Lean formalization and describe its audit procedure in §3; this literature update reviewed the paper’s statements and scope but did not rerun Lean or audit the formalization.